

Take home final examination

Course: **CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS**
Term: Spring 2020
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Due Date: Monday May 4th 2020 by 11:59 pm

Number of points.....100

Your score.....

INSTRUCTIONS FOR FINAL EXAMINATION DELIVERY

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Note: All problems are worth 100 points. Your final score will be the average value

Problem 1: Tank Pressurization Calculations with the Redlich-Kwong EOS

A cylindrical tank with volume $\{V = 25\text{ m}^3\}$ is originally filled with nitrogen at conditions of $\{T_o = 25^\circ\text{C}$ and $P_o = 1\text{ bar}\}$. The tank must be pressurized to a pressure of $\{P = 25\text{ bar}\}$ using a high pressure line which supplies nitrogen at a constant temperature and pressure equal to $\{T = 25^\circ\text{C}$ and $P = 50\text{ bar}\}$.

A. Prove that by combining the dynamic energy and mass balances we obtain the equation:

$d(MU) = H dM$ where M is the mass(in moles) of nitrogen inside the tank at any time, U is the molar internal energy of nitrogen inside the tank and H is the molar enthalpy of the nitrogen in the high pressure line feeding the tank

B. Integrate the equation obtained above and calculate the amount of nitrogen added to the tank during the pressurization process using (a) the ideal gas law and (b) the Redlich-Kwong equation of state. What is the error in assuming that the gas is ideal.

1a

[Start with unsteady macroscopic mass balance and unsteady state macroscopic energy balance]

$$\frac{dM}{dt} = W_1$$
$$\frac{d(MU)}{dt} = W_1 H_1 \quad \text{Kinetic and Potential Energies? - 2 pts}$$

[Solve for dM in the mass balance]

$$dM = W_1 dt$$

[For the energy balance get dt and W on the same side]

$$d(MU) = W_1 H_1 dt$$

[Substitute $W dt$ for dM in the energy balance from the mass balance]

$$d(MU) = W_1 dM$$

[Integrate both sides with respect to mass]

$$\int_{m_0}^m d(MU) = \int_{m_0}^m W_1 dM \quad \text{Wrong limits -2pts}$$

[Expand]

$$mU - m_0 U_0 = H_1 M - H_1 M_0$$

1b.

Ideal Gas

Using the CHE 301 calculator I can calculate the compressibility factor, enthalpy, density and internal energy.

[The compressibility factor can be calculated using the “Zcomp” function as we have the pressure, temperature and the component. But for an Ideal gas we know for a gas its always 1]

[All the Enthalpies can be found using the “Hig”. H_1 is the enthalpy of high pressure line is zero for an Ideal gas]

[Internal energy is just the enthalpy minus the work.]

$$U = H - W$$

$$W = PV$$

[Turning pressure and specific volume into Kj per Kmol]

$$\left(\frac{\text{bar} * m^3}{\text{kmol}}\right) \left(\frac{1000L}{m^3}\right) \left(\frac{100 J}{1 L * bar}\right) \left(\frac{1KJ}{1000J}\right) = \frac{100Kj}{Kmol}$$

[Starting equation]

$$mU - m_0U_0 = H_1M - H_1M_0$$

[H_1 is zero]

$$mU = m_0U_0$$

$$m = \rho * V$$

[Substitute]

$$\rho U = \rho_0 U_0$$

[Starting Values]

ideal	Temp1	25.0000	Temp1	25.0000
	Pressure1	1.0000	Pressure1	25.0000
	Z	1.0000	Z	1.0000
	H	0.0000	H	0.0000
	Specific Vol1	24.7897	Specific Vol2	0.9916
	Density1	0.0403	Density2	1.0085
	Uo	-2478.9682	Uo	-2478.9682
	Rho1*Uo	-100.0000	Rho2*U	-2500.0000

[Using extended Secant, I set “Rho1*Uo” and “Rho2*U” equal to each other by changing T2]

ideal	Temp1	25.0000	Temp1	137.1450
	Pressure1	1.0000	Pressure1	25.0000
	Z	1.0000	Z	1.0000
	H	0.0000	H	3274.9416
	Specific Vol1	24.7897	Specific Vol2	1.3646
	Density1	0.0403	Density2	0.7328
	Uo	-2478.9682	Uo	-136.4559
	Rho1*Uo	-100.0000	Rho2*U	-100.0000

$$T = 137.145^{\circ} \text{C} \quad \text{Correct}$$

[Density times the volume the container will yield us the kmols of nitrogen add]

18.3209 Kmols of N_2 added

Wrong calculation.
Answer = 17.31 kmols (-2 pts)

Redlich-Kwong

Using Redlich-Kwong much of the previously calculated parameters are the same or similar. Z is not always equal to 1. Enthalpy is uses the “enthalpy” function instead of “Hig”. This time H_1 is a value other than zero.

$$mU - m_0U_0 = H_1M - H_1M_0$$

[Using excel I calculated H_1 using the calculator at 25°C and 50 bar]

$$H_1 = -300 \quad \text{Unit? -1pt}$$

Then I made two cells, one for the left side of the equation and one the right side of the question.

Redlich-Kwong	Temp1	25.0000	Temp1	25.0000	Temp2	25.0000
	Pressure1	1.0000	Pressure1	25.0000	Pressure2	50.0000
	Z	0.9996	Z	0.9920	Z2	0.9874
	H	-6.4120	H	-155.6219	H2	-300.3600
	Specific Vol1	24.7801	Specific Vol2	0.9836		
	Density1	0.0404	Density2	1.0166	volume	0.4895
	Uo	-2484.4259	Uo	-2614.6965	U	-2748.0837
	Rho1*Uo	-100.2588	Rho2*U	-2658.2119		
	H_1					
Left		2557.9531	Right			-293.2378

[Using Extended Secant Method, I set them equal by changing the temperature]

Redlich-Kwong	Temp1	25.0000	Temp1	154.0481	Temp2	25.0000
	Pressure1	1.0000	Pressure1	25.0000	Pressure2	50.0000
	Z	0.9996	Z	1.0045	Z2	0.9874
	H	-6.4120	H	3708.0056	H2	-300.3600
	Specific Vol1	24.7801	Specific Vol2	1.4272		
	Density1	0.0404	Density2	0.7007	volume	0.4895
	Uo	-2484.4259	Uo	139.9700	U	-2748.0837
	Rho1*Uo	-100.2588	Rho2*U	98.0722		
	H_1					
Left	-198.3309	Right	-198.3309			

$$T = 154.0481^{\circ} \text{ C}$$

[Density times the volume the container will yield us the kmols of nitrogen add for Redlich Kwong]

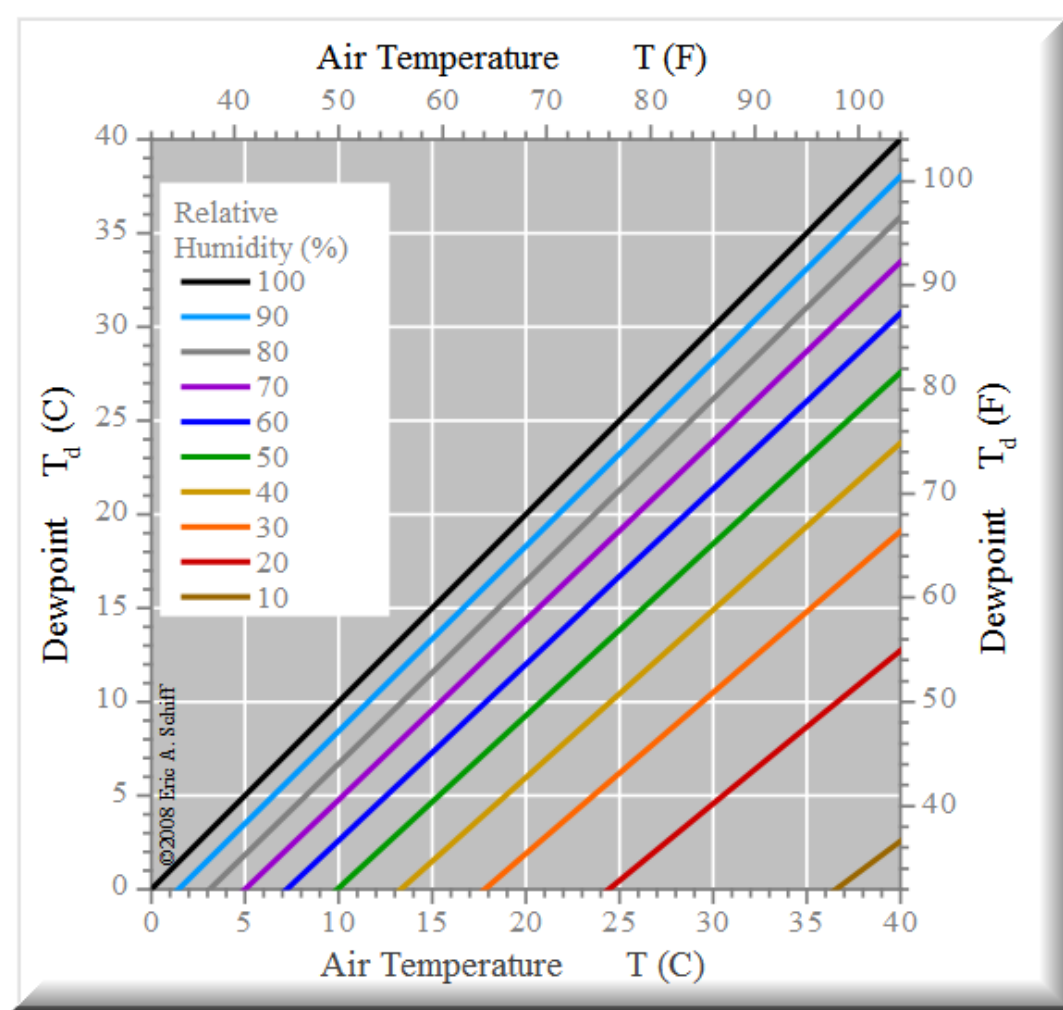
17.5166 Kmols of N₂ added

Calculation error.
 Answer = 17.68 kmols (-2 pts)
 R-K should be > IG (-2 pts)

What is the error in assuming that the gas is ideal? -5 pts

Problem 2: Dew Point Calculations

On Tuesday July 8, 2003 in Dhahran Saudi Arabia, the recorded temperature at 12:00 noon was 104°F and the sea level pressure 29.39 inHg. The relative humidity was recorded at 72%. Estimate the dew point of the air from the chart given below, and verify your estimation using a thermodynamic model of your choice. If you decide to use Henry's law for oxygen, nitrogen, argon and carbon dioxide values for the Henry's law constants for these gases can be found in your notes. The composition of dry air can be found in the open literature.



[Starting Pressure and Temperature]

$$P = 29.39 \text{ inHg} = .9952 \text{ bar}$$

$$T = 40^\circ \text{ C}$$

[The Molar Composition of Air Dry from Engineering Toolbox is the following] ok

$$N_2 = 78.080\%$$

$$O_2 = 20.95\%$$

$$Ar = .93\%$$

$$CO_2 = .033\%$$

Now we need to find the amount of water vapor in the air. We can do this by using the definition of relative humidity. Relative humidity is how much water vapor is the air as a percentage of the amount needed for saturation at a given temperature. Then normalize the composition back to one once we know the total composition.

[Definition of Relative humidity]

$$\text{Relative Humidity} = \frac{\text{Actual Vapor Pressure}}{\text{Saturated Vapor Pressure}} * 100 \quad \text{ok}$$

[Define AVP in terms of the problem]

$$\text{Actual Vapor Pressure} = \text{Pressure} * (\text{percentage of water})$$

[Substitute]

$$\text{Relative Humidity} = \frac{\text{Pressure} * H_2O}{\text{Saturated Vapor Pressure}} * 100$$

Knowing that relative Humidity is 72%, pressure, Saturated Vapor Pressure (calculated using the pSat function from the 301 calculated as we know the temperature). Then we can goal seek the right side of the equation equal to the left side which is 72 by solving for the percentage of H2O in the air.

[The result]

$$H_2O = 5.387\% \quad \text{ok}$$

If we sum all the molar compositions, (Z_i) we a number greater than 1. We can then divide each of the molar compositions by the sum. So when you sum them again they all add to 1. Should keep the calculated value for water -2pts

[New compositions]

$$N_2 = 74.094\%$$

$$O_2 = 19.88\%$$

$$Ar = .883\%$$

$$CO_2 = .031\%$$

$$H_2O = 5.112\%$$

To find the dewpoint we need to use the Richford-Rice equation and set the Alpha to one. First, we need to find the K values of the components. ok

[Finding K]

$$K = \frac{\text{Saturated Pressure}}{\text{Pressure}}$$

[K Values]

$$K_{N_2} = 1688$$

$$K_{O_2} = 2466$$

$$K_{Ar} = 871$$

$$K_{CO_2} = 69$$

$$K_{H_2O} = .074$$

[Using these values, Alpha = 1, and goal seeking the Richford-Rice equation to zero by changing temperature] ok

$$\text{Temp} = 33.0036^\circ \text{C}$$

[Using a Henry's Laws would be give a better approximation of the temperature. Henry's law uses a constant for a component in which you divide by the pressure to get a K value. For water we do not need to use Henry's Law] why? -2pts

$$K_i = \frac{H_i}{P}$$

[K Values using Henry's Law]

$$K_{N_2} = 88072$$

$$K_{O_2} = 44594$$

$$K_{Ar} = 40192$$

$$K_{CO_2} = 54863$$

$$K_{\text{H}_2\text{O}} = .074$$

[New Temperature using Henry's K values]

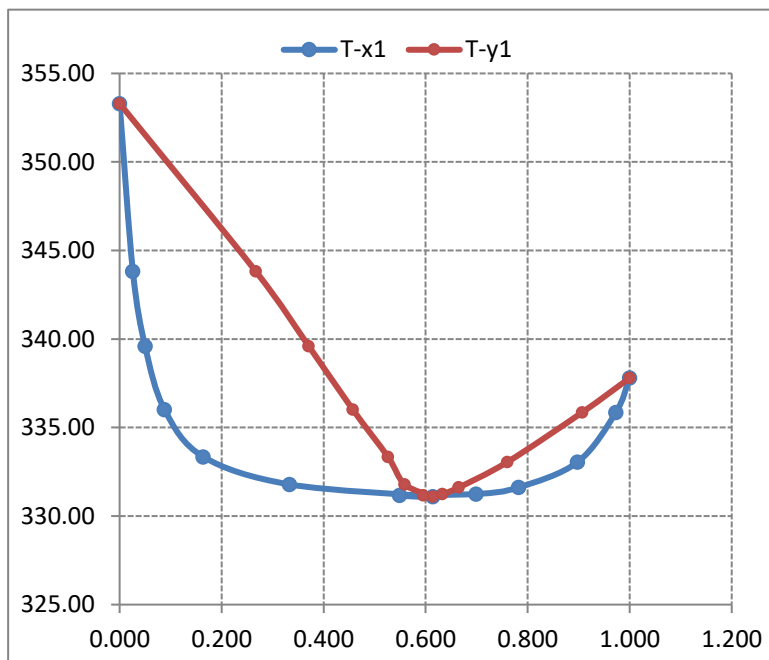
$$T_{\text{final}} = 32.992^\circ \text{C}$$

Problem 3: VLE Data Reduction with the Peng-Robinson Equation of State

Consider a mixture of Methanol (Component #1) and Benzene (Component #2) at 1.01325 bar that follows the Peng-Robinson Equation of State.

- Assume that the binary interaction parameter k_{12} is zero. Calculate the bubble point temperature at values of x_1 shown below and the dew point temperature at values of y_1 shown below. Plot the bubble point temperatures vs. liquid mole fraction x_1 and dew point temperature vs. vapor mole fraction y_1 . Superimpose the experimental data given below onto your graph and comment on the goodness of fit, by calculating the mean residual sum of squares. A value less than 10.0 indicates an acceptable result.
- At the azeotrope, the composition of liquid and vapor phases are the same at equilibrium. Use trial and error to determine the temperature and composition that is predicted for the azeotrope between methanol and benzene at 1.01325 bar if k_{12} is zero. Compare your result with the actual values given in the data below.
- Determine the optimal value of the binary interaction parameter k_{12} that can be used for the prediction of the azeotrope of this mixture

Temp(C)	x_1	x_2	y_1	y_2
80.15	0.000	1.000	0.000	1.000
70.67	0.026	0.974	0.267	0.733
66.44	0.050	0.950	0.371	0.629
62.87	0.088	0.912	0.457	0.543
60.20	0.164	0.836	0.526	0.474
58.64	0.333	0.667	0.559	0.441
57.95	0.614	0.386	0.614	0.386
58.02	0.549	0.451	0.595	0.405
58.10	0.699	0.301	0.633	0.367
58.47	0.782	0.218	0.665	0.335
59.90	0.898	0.102	0.760	0.240
62.71	0.973	0.027	0.907	0.093
64.65	1.000	0.000	1.000	0.000



3a.

Find the Bubble point temperature given X_i

Given the starting pressure and the molar composition of the mixture we need to find the temperature of the Dew Point and Bubble Point. Meaning we need to find the K values first in order to solve the Richford-Rice equation to zero. Also, the molar composition of the mixture is equal to the liquid molar composition.

$$Z_i = X_i$$

We can find the starting K value by knowing that it is the pSat divided by the pressure.

$$K_i = \frac{p_{\text{Sat}}}{P}$$

We can find the pSat by using the function “Psat” given that we have temperature and pressure. Then using the initial guess from pSat we can use “EOSPAT” as we are using the Peng-Robinson equation of state. We only use these K for an initial guess. After we get a fugacity we change the way we get K values by relating it to fugacity.

[Relating K values and fugacity]

$$K_i = \frac{\hat{\Phi}_i^L}{\hat{\Phi}_i^V}$$

The fugacity coefficient can be obtained using the “phiC” function in the Che 311 calculator. Requiring the Y_i , the component, T, P, and Z. the Z.

The image shows two side-by-side screenshots of a software interface for calculating the Fugacity Coefficient (PhiC). Each interface has a table of input parameters and a result field.

Parameter	Value	Result
MolFrac	y1	[0.109101873815318, 0.890898347...
CompCol	comp1	"CH4O"
T	Temp	77.85446481
P	Pres	1.01325
Z	B19	0.971767061
		= 0.985654913

Calculates the Fugacity Coefficient.

MolFrac Molar Composition Range.

Formula result = 0.98565

[Help on this function](#) [OK] [Cancel]

Parameter	Value	Result
MolFrac	0.05, 0.95	[0.05, 0.95]
CompCol	comp1	"CH4O"
T	Temp	77.85446481
P	Pres	1.01325
Z	B30	0.003147844
		= 2.150735959

Calculates the Fugacity Coefficient.

MolFrac Molar Composition Range.

Formula result = 2.15074

[Help on this function](#) [OK] [Cancel]

The Z was calculated using the mixture not an individual component, so the composition was X_i and Y_i respectively with the same temperature and pressure. The initial guess was 1 for a vapor and 0 for a liquid. Now that we have all fugacity of the liquid and vapor, we can get a K value for both components. Setting the vapor feed ratio or alpha to zero will yield us the bubble point.

Starting values

Fluid Conditions	Mixture	CH4O	C6H6	
Molar Composition	1.0000	0.0500	0.95000	
Temperature, C	80.1400			
Pressure, bar	1.013250			
Saturated Vapor Pressure, bar		1.8655	1.0163	1.74233
Saturated Vapor Temperature, C				65.53196
K-Values		2.35128	1.00108	
New K-Values		2.35128	1.00108	
Vapor to Feed Molar Ratio [a=V/F]	0.0000			
The Rachford-Rice Equation	0.0686			
Discriminant of the Cubic EOS				
Vapor Phase				
Molar Composition		1.00000	0.00000	
New Molar Composition		0.11756	0.95103	
Compressibility Factor	0.9840			
Fugacity Coefficient		0.98421	0.97347	
Saturated Fugacity Coefficient				

After solving

Fluid Conditions	Mixture	CH4O	C6H6	
Molar Composition	1.0000	0.0500	0.95000	
Temperature, C	77.8545			
Pressure, bar	1.013250			
Saturated Vapor Pressure, bar		1.7162	0.9479	1.60662
Saturated Vapor Temperature, C				65.53196
K-Values		2.18204	0.93779	
New K-Values		2.18204	0.93779	
Vapor to Feed Molar Ratio [a=V/F]	0.0000			
The Rachford-Rice Equation	0.0000			
Discriminant of the Cubic EOS				
Vapor Phase				
Molar Composition		0.10910	0.89090	
New Molar Composition		0.10910	0.89090	
Compressibility Factor	0.9718			
Fugacity Coefficient		0.98565	0.97088	

for the vapor phase, I guessed that methanol is going to be 100% of the vapor composition because it is the ore volatile component. Use CHE 301 Calculator function “Bubble Point Temperature”, getting a value for the bubble point temperature for that vapor composition. So, we’ll have to update the vapor composition. The new vapor composition is just the K value time the Zi.

$$Y_i^{\text{new}} = Z_i * K_i$$

Putting that equation in the row “New Molar Composition” and hitting “Update Y-Values” replaces the current guess with a better one. Solve for a better bubble point and a use the updated Y-values until you converge. Once convergence happens that is the final molar composition and bubble point temperature.

Repeat that process to find the Y_i and bubble point temperatures for all compositions

Temp	X1	X2	Y1	Y2
80.1500	0.0000	1.0000	0.0000	1.0000
78.8614	0.0260	0.9740	0.05922	0.9408
77.8249	0.0500	0.9500	0.11020	0.8898
76.2746	0.0880	0.9120	0.18456	0.8154
73.7000	0.1640	0.8360	0.30311	0.6969
69.5988	0.3330	0.6670	0.5021	0.4979
65.6080	0.6140	0.3860	0.7182	0.2818
66.2788	0.5490	0.4510	0.6755	0.3245
64.9166	0.6990	0.3010	0.7713	0.2287
64.4278	0.7820	0.2180	0.8228	0.1772
64.0671	0.8980	0.1020	0.9031	0.0969
64.1071	0.9730	0.0270	0.9701	0.0299
64.2038	1.0000	0.0000	1.0000	0.0000

Find the Dew point temperature given Y_i

Many of the steps for finding the dew point are the same as the finding the bubble point. The fugacity coefficient, compressibility factor, saturated pressure, and pressure are the same as above. The α is equal to 1 and not 0 this time. The Z_i is also equal to the Y_i not the X_i .

$$Z_i = Y_i$$

[A guess will be entered for the molar composition of the liquid phase, and the new molar composition will be equal to the following for.]

$$X_i^{\text{new}} = \frac{Y_i}{K_i}$$

Fluid Conditions	Mixture	CH4O	C6H6	
Molar Composition	1.0000	0.6140	0.38600	
Temperature, C	80.0406			
Pressure, bar	1.013250			
Saturated Vapor Pressure, bar		1.8588	1.0133	1.73625
Saturated Vapor Temperature, C				65.53196
K-Values		2.37537	0.99917	
New K-Values		2.37537	0.99917	
Vapor to Feed Molar Ratio [a=V/F]	1.0000			
The Rachford-Rice Equation	0.3552			
Discriminant of the Cubic EOS				
Vapor Phase				
Molar Composition	1.0000	0.6140	0.38600	
New Molar Composition		1.45848	0.38568	
Compressibility Factor	0.9793			
Fugacity Coefficient		0.98452	0.97213	
Saturated Fugacity Coefficient				
Molar Enthalpy, kJ/kmol				
Molar Entropy, kJ/(kmol K)				
Molar Density, kmol/m ³				
Molar Heat Capacity, kJ/(kmol K)				
Liquid Phase				
Molar Composition	1.0000	0.00000	1.00000	
New Molar Composition		0.25849	0.38632	
Compressibility Factor	0.0032	0.00321	0.00177	
Fugacity Coefficient		2.33860	0.97132	

Using the 301 function calculate “Dew Point Temperature” which solves the Richford-Rice equation to zero by finding a new temperature value that satisfies the condition. Then we iterate the new molar composition using “Update X-Vales”, iterate until we converge onto a value.

[New values after process]

Fluid Conditions	Mixture	CH ₄ O	C ₆ H ₆	
Molar Composition	1.0000	0.6140	0.38600	
Temperature, °C	67.3901			
Pressure, bar	1.013250			
Saturated Vapor Pressure, bar		1.1512	0.6796	1.08909
Saturated Vapor Temperature, °C				65.53196
K-Values		1.32511	0.71929	
New K-Values		1.32511	0.71929	
Vapor to Feed Molar Ratio [a=V/F]	1.0000			
The Rachford-Rice Equation	0.0000			
Discriminant of the Cubic EOS				
Vapor Phase				
Molar Composition	1.0000	0.61400	0.38600	
New Molar Composition		0.81362	0.27765	
Compressibility Factor	0.9771			
Fugacity Coefficient		0.98275	0.96933	
Saturated Fugacity Coefficient				
Molar Enthalpy, kJ/kmol				
Molar Entropy, kJ/(kmol K)				
Molar Density, kmol/m ³				
Molar Heat Capacity, kJ/(kmol K)				
Liquid Phase				
Molar Composition	1.0000	0.46336	0.53664	
New Molar Composition		0.46336	0.53664	
Compressibility Factor	0.0026	0.00327	0.00180	
Fugacity Coefficient		1.30225	0.69723	
Liquid Activity Coefficient				

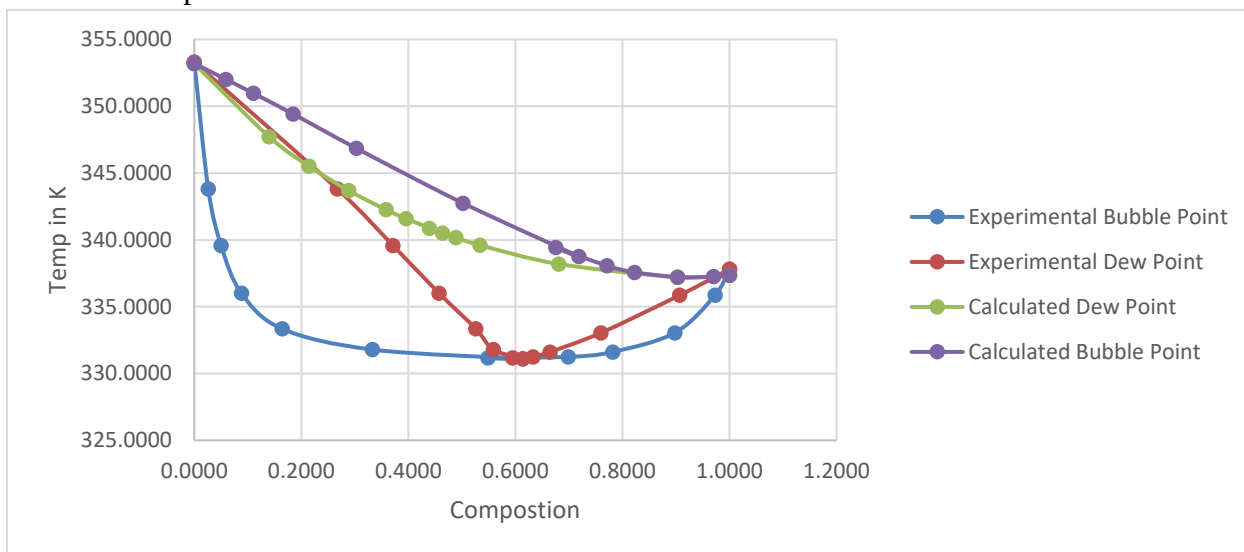
[Repeat this step for the remaining mol fractions. Here are the dew point temperatures]

Temp	X1	X2	Y1	Y2
80.0406	0.0000	1.0000	0.0000	1.0000
74.5719	0.1397	0.8603	0.2670	0.7330
72.3651	0.2143	0.7857	0.3710	0.6290
70.5414	0.2885	0.7115	0.4570	0.5430
69.1067	0.3584	0.6417	0.5260	0.4740
68.4395	0.3955	0.6045	0.5590	0.4410
67.3721	0.4635	0.5365	0.6140	0.3860
67.7333	0.4392	0.5608	0.5950	0.4050
67.0209	0.4889	0.5111	0.6330	0.3670
66.4551	0.5337	0.4663	0.6650	0.3350
65.0487	0.6807	0.3193	0.7600	0.2400
64.0617	0.9029	0.0971	0.9070	0.0930
64.2037	1.0000	0.0000	1.0000	0.0000

Doing the Residual Sum of the data shows that our model is not a good fit as the average for both of them are much greater than 10.

BubblePoint			(Data-Calc)*2			BubblePoint			(Data-Calc)*2		
X1	Temp From Data	Calculated Temp	Error	Average		Y1	Temp From Data	Calculated Temp	Error	Average	
0.0000	80.1500	80.0460	0.0108	69.7759		0.0000	80.1500	80.0406	0.0120	49.1966	
0.0260	70.6700	78.8614	67.0998			0.2670	70.6700	74.5719	15.2246		
0.0500	66.4400	77.8249	129.6159			0.3710	66.4400	72.3651	35.1065		
0.0880	62.8700	76.2746	179.6845			0.4570	62.8700	70.5414	58.8501		
0.1640	60.2000	73.7000	182.2500			0.5260	60.2000	69.1067	79.3293		
0.3330	58.6400	69.5988	120.0957			0.5590	58.6400	68.4395	96.0306		
0.6140	57.9500	65.6080	58.6453			0.6140	57.9500	67.3721	88.7760		
0.5490	58.0200	66.2788	68.2073			0.5950	58.0200	67.7333	94.3475		
0.6990	58.1000	64.9166	46.4662			0.6330	58.1000	67.0209	79.5816		
0.7820	58.4700	64.4278	35.4957			0.6650	58.4700	66.4551	63.7617		
0.8980	59.9000	64.0671	17.3648			0.7600	59.9000	65.0487	26.5093		
0.9730	62.7100	64.1071	1.9520			0.9070	62.7100	64.0617	1.8271		
1.0000	64.6500	64.2038	0.1991			1.0000	64.6500	64.2037	0.1992		

Here is the experimental data vs our calculated data.



3b.

To figure out the azeotrope we need to find out the molar composition and the temperature it forms at. Because the vapor and liquid are at equilibrium, we can set the alpha to any number between zero and one as it is . Use the liquid composition as the total composition. The total molar composition must sum to 1.

$$\alpha = 0$$

$$Z_i = X_i$$

$$\sum Z_i = 1$$

Everything else such as the fugacity coefficient, Z, and pSat will be the same as above.

The key idea of the problem is that the vapor and liquid are in equilibrium. That means that the fugacity of the vapor and the liquid are equal.

$$\hat{\Phi}_i^V = \hat{\Phi}_i^L$$

For me to find the correct composition and temperature I need to have accurate starting composition. Looking at the experimental graph on 3a, it looks like the 2 calculated lines of the bubble point and dew point meet around 90% methanol and 10% benzene. Using goal, I set the Rachford-Rice equation to zero.

These are my starting numbers on everything

Fluid Conditions	Mixture	CH4O	C6H6
Molar Composition	1.0000	0.9000	0.10000
Temperature, C	80.0000		
Pressure, bar	1.013250		
Saturated Vapor Pressure, bar		1.8561	1.0120
Saturated Vapor Temperature, C			
K-Values		1.82085	1.43828
New K-Values		1.82085	1.43828
Vapor to Feed Molar Ratio [a=V/F]	0.0000		
The Rachford-Rice Equation	0.7826		
Discriminant of the Cubic EOS			
Vapor Phase			
Molar Composition	1.0000	1.00000	0.00000
New Molar Composition		1.63876	0.14383
Compressibility Factor	0.9840		
Fugacity Coefficient		0.98419	0.97344
Saturated Fugacity Coefficient			
Molar Enthalpy, kJ/kmol			
Molar Entropy, kJ/(kmol K)			
Molar Density, kmol/m3			
Molar Heat Capacity, kJ/(kmol K)			
Liquid Phase			
Molar Composition	1.0000	0.90000	0.10000
New Molar Composition		0.54919	0.00000
Compressibility Factor	0.0019	0.00321	0.00177
Fugacity Coefficient		1.79207	1.40008
Liquid Activity Coefficient			

After Goal Seek, this temperature makes sense as my graph above shows that the temperature should be between 62° C and 67° C. So, this would be a valid answer.

$$T = 64.0704^{\circ} \text{ C}$$

Then we need to set the fugacity of the vapor and liquid equal. By using the Extended Secant Method, I set the fugacity of the methanol of vapor equal to the liquid by changing the molar composition of methanol.

Before	$\hat{\Phi}_1^V = \hat{\Phi}_1^L$,	$\hat{\Phi}_1^V = .98190$,	$\hat{\Phi}_1^L = .98725$,	$X_1 = .9$
After	$\hat{\Phi}_1^V = \hat{\Phi}_1^L$,	$\hat{\Phi}_1^V = .98190$,	$\hat{\Phi}_1^L = .98190$,	$X_1 = .9316$

Then set the fugacity of the benzene equal to each other using extended Secant method by changing X_1 . It can only work if your benzene composition is related to composition of methanol.

		$X_2 = (1 - X_1)$		
Before	$\hat{\Phi}_1^V = \hat{\Phi}_1^L$,	$\hat{\Phi}_1^V = .96996$,	$\hat{\Phi}_1^L = .97871$,	$X_1 = .9316$
After	$\hat{\Phi}_1^V = \hat{\Phi}_1^L$,	$\hat{\Phi}_1^V = .96996$,	$\hat{\Phi}_1^L = .96996$,	$X_1 = .9270$

$$X_2 = .07297$$

Because the composition changed, we will need to set the Richford-Rice Equation equal to zero. Using goal seek once again we get a small about change.

$$T = 64.0539^\circ \text{C}$$

As this might take a couple of tries to get everything equal to zero. So it would be better to use solver to make this process more efficient. These would be the following requirements by changing temperature, X1 and X2.

- $\sum Z_i = 1$
- Richford-Rice = 0
- $\hat{\Phi}_1^V = \hat{\Phi}_1^L$
- $\hat{\Phi}_2^V = \hat{\Phi}_2^L$

The objective function would the sum(Z_i) =1 while the other 3 would be the constraints.

Set Objective:

To: ☐ Max ☐ Min ☒ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

-
-
-

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Buttons: Add, Change, Delete, Reset All, Load/Save, Options, Help, Solve, Close

Using solver and our initial conditions we get these as our final answer.

Fluid Conditions	Mixture	CH4O	C6H6
Molar Composition	1.0000	0.9273	0.07274
Temperature, C	64.0541		
Pressure, bar	1.013250		
Saturated Vapor Pressure, bar		1.0071	0.6082
Saturated Vapor Temperature, C			
K-Values		1.00000	1.00000
New K-Values		1.00000	1.00000
Vapor to Feed Molar Ratio [a=V/F]	0.6654		
The Rachford-Rice Equation	0.0000		
Discriminant of the Cubic EOS			
Vapor Phase			
Molar Composition	1.0000	1.00000	0.00000
New Molar Composition		0.92726	0.07274
Compressibility Factor	0.9816		
Fugacity Coefficient		0.98190	0.96995
Saturated Fugacity Coefficient			
Molar Enthalpy, kJ/kmol			
Molar Entropy, kJ/(kmol K)			
Molar Density, kmol/m ³			
Molar Heat Capacity, kJ/(kmol K)			
Liquid Phase			
Molar Composition	1.0000	0.92726	0.07274
New Molar Composition		1.00000	0.00000
Compressibility Factor	0.0019	0.00329	0.00181
Fugacity Coefficient		0.98190	0.96995

My results

$$T = 64.0541, \quad X_1 = .9274, \quad X_2 = .07274$$

Experimental data

$$T = 57.95, \quad X_1 = .614, \quad X_2 = .386$$

Once again, our model does is not good at predicting the actual data. Being off by 6 K and the molar composition is not remotely close to the real.

3c.

The binary interaction parameter is a measure of how unideal a mixture is. To make sure we get a good K12 value we need to get a temperature and molar composition of close to the experimental case. So, a way to find out how much error there is, we can use a residual error.

$$R_1 = (X_{\text{Exp}} - X_1)^2$$

$$R_2 = \frac{(T_{\text{Exp}} - T)^2}{100}$$

I added a K12 as a parameter in my database just was just in my Zcomp function which would help would figuring out how unideal the mixture is.

Fluid Conditions	Mixture	CH4O	C6H6		
Molar Composition	1.0000	0.6000	0.40000		
Temperature, C	58.0000				
Pressure, bar	1.013250				
Saturated Vapor Pressure, bar		0.7835	0.4941	0.74848	0.47805
Saturated Vapor Temperature, C				65.53196	81.01762
K-Values		4.14394	3.77669	R1	0.0002
New K-Values		4.14394	3.77669	R2	0.0000
				R1+R2	0.0002
Vapor to Feed Molar Ratio [a=V/F]	0.0000			Kij	0.5000
The Rachford-Rice Equation	2.9970			Zv	0.9819
Discriminant of the Cubic EOS				Zl	0.0027
Vapor Phase					
Molar Composition	1.0000	0.60000	0.40000		
New Molar Composition		2.48636	1.51067		
Compressibility Factor	0.9819				
Fugacity Coefficient		0.98569	0.97683		
Saturated Fugacity Coefficient					
Molar Enthalpy, kJ/kmol					
Molar Entropy, kJ/(kmol K)					
Molar Density, kmol/m3					
Molar Heat Capacity, kJ/(kmol K)					
Liquid Phase					
Molar Composition	1.0000	0.60000	0.40000		
New Molar Composition		0.14479	0.10591		
Compressibility Factor	0.0027	0.00333	0.00183		
Fugacity Coefficient		4.08464	3.68919		
Liquid Activity Coefficient					

I then used solver to try get a value for X1, X2, K12, and temp by using them as a variable. Having my R1+R2 as my objective function which was set to a minimum. The constraints would be the Richford-Rice equation to zero and both of the fugacities of the vapor and liquid equal to each other.

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

When I first tried to use my solver, I couldn't convergence on to a value. So I used extended secant method to set the liquid fugacity and vapor fugacity of the benzene equal to each other by changing K12, which ended up being .15. Then solver converged.

Fluid Conditions	Mixture	CH ₄ O	C ₆ H ₆		
Molar Composition	1.0000	0.6932	0.30682		
Temperature, C	57.8537				
Pressure, bar	1.013250				
Saturated Vapor Pressure, bar		0.7786	0.4916	0.74396	0.47358
Saturated Vapor Temperature, C				65.53196	81.01762
K-Values		1.00000	1.00000	R1	0.0063
New K-Values		1.00000	1.00000	R2	0.0001
				R1+R2	0.0064
				Kij	0.0847
Vapor to Feed Molar Ratio [a=V/F]	0.0000				
The Rachford-Rice Equation	0.0000			Zv	0.9774
Discriminant of the Cubic EOS				Zl	0.0023
Vapor Phase					
Molar Composition	1.0000	0.69318	0.30682		
New Molar Composition		0.69318	0.30682		
Compressibility Factor	0.9774				
Fugacity Coefficient		0.98156	0.96944		
Saturated Fugacity Coefficient					
Molar Enthalpy, kJ/kmol					
Molar Entropy, kJ/(kmol K)					
Molar Density, kmol/m ³					
Molar Heat Capacity, kJ/(kmol K)					
Liquid Phase					
Molar Composition	1.0000	0.69318	0.30682		
New Molar Composition		0.69318	0.30682		
Compressibility Factor	0.0023	0.00333	0.00183		
Fugacity Coefficient		0.98156	0.96944		

[Final Answers after solver]

$$T = 57.8537^{\circ} \text{C}, \quad K_{ij} = .0847, \quad X_1 = .6932, \quad X_2 = .30682$$

Problem 4: Excess Enthalpy Model for Isopropanol-Water Mixtures

Your first project at the XYZ corporation is to develop a methodology for the calculation of the molar enthalpy of isopropyl alcohol(1)-water(2) liquid mixtures at different temperatures and pressure equal to 1 atm. There is a large number of the excess enthalpy experimental data available at different temperatures and mixture compositions and these are given in the table below: (all data are atmospheric pressure)

T=273.15K x_1	H^E , kJ/kmol	T=308.15K x_1	H^E , kJ/kmol	T=328.15K x_1	H^E , kJ/kmol
0.001	-20.934	0.0029	33.494	0.0047	-36.006
0.005	-97.134	0.013	-121.417	0.019	-115.137
0.014	-257.907	0.057	-355.878	0.063	-209.34
0.026	-439.614	0.103	-456.361	0.084	-234.461
0.083	-866.667	0.151	-439.614	0.1	-247.021
0.087	-870.854	0.351	-171.659	0.138	-198.873
0.135	-983.898	0.448	-33.494	0.158	-173.333
0.179	-921.096	0.577	157.424	0.163	-160.354
0.226	-862.481	0.673	211.015	0.25	-8.792
0.509	-401.933	0.767	217.714	0.359	146.538
0.704	-106.345	0.891	146.538	0.456	292.657
0.788	-9.211	0.929	103.833	0.615	410.306
0.864	33.494	0.969	48.148	0.637	418.68
0.904	30.982			0.789	319.871
0.939	22.609			0.903	185.057
				0.909	174.59
				0.967	69.92

- Develop a model for the Excess Enthalpy H^E using the thermodynamic principles you have learnt in class and estimate the parameters of the model from the experimental data.
- Using the model you developed for the excess enthalpy calculate the amount of heat in kJ per kmol of solution exchanged with the surroundings when 20 moles of isopropyl alcohol are mixed **isothermally** with 80 moles of water at 315.15 °K.
- If the pure components are initially at 315.15 °K and the mixing is **adiabatic** what is the final temperature of the solution.

4a.

[Calculate X_2 after inputting all data]

$$X_2 = (1 - X_1)$$

To create the model we need to get 3 constants for the second order regression. To get these 3 constants we need to have 3 sets of data to get 3 constants. These constants can be found by the following.

[First]

$$X_1 X_2$$

[Second]

$$X_1^2 X_2$$

[Third]

$$X_1^3 X_2$$

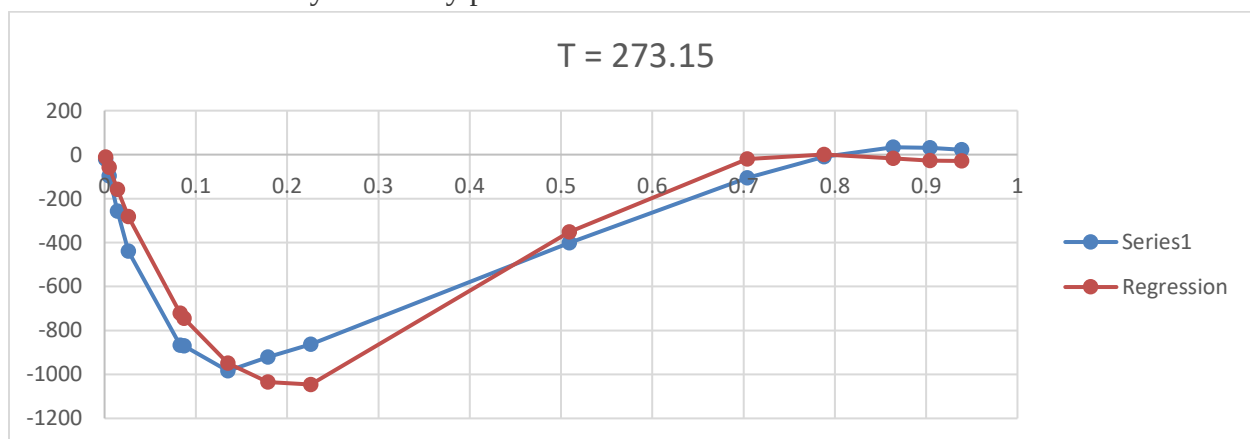
Make 3 columns using the 3 equations above. Using the excel built in tool “Data Analysis” we can use the regression function. After highlighting the 3 columns above and the HE data we get 3 constants.

$$\text{At } T = 273.15 \text{ K, } A_0 = -11900, \quad A_1 = 30664, \quad A_2 = -19744 \quad \text{ok}$$

These constants can be used to make an equation to model the excess enthalpy at temperature T at a composition. This is the following equation.

$$H^E = X_1 X_2 (A_0 + A_1 X_1 + A_2 X_1^2) \quad \text{ok}$$

I then made a column calculating the excess enthalpy from the regression and plot the results with the original data given. The Regression is a decent fit as most of the points are close to the real data as well at many of the key points so are to each other.



[Do the previous steps for the other temperatures to calculate coefficients for the regression]

At T = 308.15 K, $A_0 = -6300$, $A_1 = 30664$, $A_2 = -19744$ Wrong values of A1 and A2 -3 pts

At T = 328.15 K, $A_0 = -3572$, $A_1 = 15183$, $A_2 = -10306$ ok

Getting decent results once again, we need to combine all of the data into one model. If we take the natural log of the Temp then graph each of the A_n then there is a linear relationship. Using the linear function which gives you a slope and y-intercept from x (ln T) and y (A_n) values.

$$\begin{aligned} A_0 &= 45502(\ln T) - 267150 \\ A_1 &= -85616(\ln T) + 510732 \quad \text{ok} \\ A_2 &= 52795(\ln T) - 315661 \end{aligned}$$

To test if my model is accurate, I set the temperature to 273.15 K, and comparing my model's results to the ones I got earlier. The results are very close to the previous constants, so this model is accurate. This model would only work between the temperature between 273.15 K and 328.15 K.

At T = 273.15 K, $A_0 = -11878$, $A_1 = 30419$, $A_2 = -19475$

4b.

To find out the heat in KJ per Kmol requires us to find the excess enthalpy. The composition is 20% isopropyl alcohol and the 80% water at a temperature of 315.15 K.

$$\begin{aligned} \bar{H} &= \bar{H}_1 + \bar{H}_2 + H^E \\ \bar{H}_1 &= X_1 H_1 \quad \text{ok} \\ \bar{H}_2 &= X_2 H_2 \end{aligned}$$

The enthalpy of the pure component can be found by using the difference of enthalpy at saturated and enthalpy of vaporization. Using the CHE 301 calculator I got the following values.

$$\begin{aligned} \bar{H}_1 &= (.2)(-314520) = -62904 \\ \bar{H}_2 &= (.8)(-285142) = -228114 \end{aligned} \quad \begin{array}{l} \text{How did you calculate H1 and H2?} \\ -3 \text{ pts} \end{array}$$

[Calculating excess enthalpy given conditions using the model]

$$\begin{aligned} X_1 &= .2, \quad X_2 = .8, \quad T = 315.15 \text{ K} \\ H^E &= -354 \quad \text{ok} \end{aligned}$$

[Plug all the values into the initial equation above to the enthalpy of the mixture]

$$\bar{H} = -291372 \frac{\text{KJ}}{\text{Kmol}}$$

Wrong answer.
Need to solve for Q
-5 pts

4c.

[Start with the energy and mass balance]

$$M_1 + M_2 = M_3$$

$$M_1 H_1 + M_2 H_2 = M_3 H_3$$

[Substitute M_3 from mass balance to energy balance]

$$M_1 H_1 + M_2 H_2 = (M_1 + M_2) H_3$$

[Rearrange]

$$M_1 H_1 + M_2 H_2 = M_1 H_3 + M_2 H_3$$

$$M_1 H_1 - M_1 H_3 = M_2 H_3 - M_2 H_2$$

$$M_1 (H_1 - H_3) = M_2 (H_3 - H_2)$$

$$\frac{M_1}{M_2} = \frac{H_3 - H_2}{H_1 - H_3}$$

[Plug in values and solve for the final enthalpy]

$$\frac{.2}{.8} = \frac{H_3 + 285142}{-314520 - H_3}$$

$$H_3 = -291018$$

Using the secant method we set H_3 and $\bar{H}$ equal to each by changing temperature which would change the excess enthalpy. $\bar{H}$ is dependent on the excess enthalpy and the temperature of both enthalpies of H_1 and H_2 .

$T = 44.997$

ok

Problem 5: VLE Data Reduction with the NRTL Model

The following **Pxy** data for a Toluene/n-Decane mixture are available at $T = 373.47^\circ\text{K}$:

TEMPERATURE	PRESSURE	X ₁	X ₂	Y ₁	Y ₂
K	Pa	TOLUENE	n-DECANE	TOLUENE	n-DECANE
373.47	9450	0.0000	1.0000	0.0000	1.0000
373.47	11630	0.0353	0.9647	0.2415	0.7585
373.47	14180	0.0700	0.9300	0.3834	0.6166
373.47	17240	0.1223	0.8777	0.5147	0.4853
373.47	22710	0.1880	0.8120	0.6544	0.3456
373.47	29530	0.3006	0.6994	0.7732	0.2268
373.47	34260	0.3665	0.6335	0.8277	0.1723
373.47	45240	0.5301	0.4699	0.9029	0.0971
373.47	50040	0.6144	0.3856	0.9251	0.0749
373.47	58680	0.7769	0.2231	0.9560	0.0440
373.47	66370	0.8617	0.1383	0.9746	0.0254
373.47	75380	1.0000	0.0000	1.0000	0.0000

From the **Gibbs Excess** energy function for the **NRTL (Non-Random Two Liquid)** model the liquid activity coefficients for the two mixture components are given by:

$$\ln \gamma_1 = x_2^2 \left[\tau_{21} \left(\frac{G_{21}}{x_1 + x_2 G_{21}} \right)^2 + \frac{\tau_{12} G_{12}}{(x_2 + x_1 G_{12})^2} \right]$$

$$\ln \gamma_2 = x_1^2 \left[\tau_{12} \left(\frac{G_{12}}{x_2 + x_1 G_{12}} \right)^2 + \frac{\tau_{21} G_{21}}{(x_1 + x_2 G_{21})^2} \right]$$

where

$$\ln G_{12} = -0.3\tau_{12} \text{ and } \ln G_{21} = -0.3\tau_{21}$$

A. Prove the following functional expressions:

$$\ln \gamma_1^\infty = \tau_{21} + \tau_{12} \exp(-0.3\tau_{12})$$

$$\ln \gamma_2^\infty = \tau_{12} + \tau_{21} \exp(-0.3\tau_{21})$$

B. Determine whether that experimental data are thermodynamically consistent.

C. Estimate from the experimental data the **NRTL** model coefficients τ_{12}, G_{12} and τ_{21}, G_{21} and draw a parity plot of predicted pressure values vs. experimental values. (Hint: Use the expressions for liquid activity coefficients at infinite dilution to obtain initial guesses for the constants)

5a.

[Proving $\ln(\gamma_1^\infty)$]

$$\ln \gamma_1 = x_2^2 \left[\tau_{21} \left(\frac{G_{21}}{x_1 + x_2 G_{21}} \right)^2 + \frac{\tau_{12} G_{12}}{(x_2 + x_1 G_{12})^2} \right]$$

$$\ln(G_{12}) = -.3\tau_{12} \quad \ln(G_{21}) = -.3\tau_{21}$$

[Solve for G_{12} and G_{21}]

$$G_{12} = e^{-.3\tau_{12}} \quad G_{21} = e^{-.3\tau_{21}}$$

[Substitute]

$$\ln \gamma_1 = x_2^2 \left[\tau_{21} \left(\frac{e^{-.3\tau_{21}}}{x_1 + x_2 e^{-.3\tau_{21}}} \right)^2 + \frac{\tau_{12} e^{-.3\tau_{12}}}{(x_2 + x_1 e^{-.3\tau_{12}})^2} \right]$$

[Assume that we are at infinite dilution of X_1 , then $X_1 = 0$ and $X_2 = 1$]

$$\ln(\gamma_1^\infty) = 1^2 \left[\tau_{21} \left(\frac{e^{-.3\tau_{21}}}{0 + 1 * e^{-.3\tau_{21}}} \right)^2 + \frac{\tau_{12} e^{-.3\tau_{12}}}{(1 + 0 * e^{-.3\tau_{12}})^2} \right]$$

[Simplify]

$$\ln(\gamma_1^\infty) = \left[\tau_{21} \left(\frac{e^{-.3\tau_{21}}}{e^{-.3\tau_{21}}} \right)^2 + \tau_{12} e^{-.3\tau_{12}} \right]$$

[Finally]

$$\boxed{\ln(\gamma_1^\infty) = \tau_{21} + \tau_{12} e^{-.3\tau_{12}}}$$

[Proving $\ln(\gamma_2^\infty)$ which is a very similar derivation as above]

$$\ln \gamma_2 = x_1^2 \left[\tau_{12} \left(\frac{G_{12}}{x_2 + x_1 G_{12}} \right)^2 + \frac{\tau_{21} G_{21}}{(x_1 + x_2 G_{21})^2} \right]$$

[Substitute for G_{12} and G_{21}]

$$\ln \gamma_2 = x_1^2 \left[\tau_{12} \left(\frac{e^{-.3\tau_{12}}}{x_2 + x_1 e^{-.3\tau_{12}}} \right)^2 + \frac{\tau_{21} e^{-.3\tau_{21}}}{(x_1 + x_2 e^{-.3\tau_{21}})^2} \right]$$

[Assume that we are at infinite dilution of X_2 , then $X_2 = 0$ and $X_1 = 1$]

$$\ln \gamma_2^\infty = 1^2 \left[\tau_{12} \left(\frac{e^{-.3\tau_{12}}}{0 + 1 * e^{-.3\tau_{12}}} \right)^2 + \frac{\tau_{21} e^{-.3\tau_{21}}}{(1 + 0 * e^{-.3\tau_{21}})^2} \right]$$

[Simplify]

$$\ln(\gamma_2^\infty) = \tau_{12} + \tau_{21} e^{-.3\tau_{21}}$$

5b.

[Thermodynamic consistency]

$$\text{Thermodynamic consistency} = \int_0^1 \ln \left(\frac{\gamma_2}{\gamma_1} \right) dX_1 = 0$$

[What gammas are equal to]

$$\gamma_1 = \left(\frac{Y_1}{X_1} \right) \left(\frac{P}{p_{\text{toluene}}^{\text{sat}}} \right)$$

$$\gamma_2 = \left(\frac{Y_2}{X_2} \right) \left(\frac{P}{p_{\text{n-Decane}}^{\text{sat}}} \right)$$

[the Psat of each compound]

$$p_{\text{toluene}}^{\text{sat}} = 75.380 \text{ KPa}$$

$$p_{\text{n-Decane}}^{\text{sat}} = 9.450 \text{ KPa}$$

[Using the trapezoidal rule, we can find the area under the curve]

	trapiziodal rule
	0.0021
	-0.0003
	-0.0007
	0.0011
	0.0011
	0.0060
	0.0004
	-0.0219
	-0.0214
sum	-0.0337

The area under the curve is less than the absolute vale of .1. So a result we can conclude that thermodynamic consistency is real with some experimental error

5c.

To make a model to calculate a pressure for a mixture we can use the “GammaC” function from the CHE 301 calculator. Inputting X1,Y1, comp1, Temperature, using the NRTL method, A12,A21,B12,B21, and C. We can get Gamma Value for both components.

Function Arguments

GammaC

MolFrac: C4:D4 = {0,1}

CompCol: comp1 = "C7H8"

T: A4 = 373.47

Method: 4 = 4

BIP: \$Q\$5:\$Q\$9 = {-0.253579639484497;0.30338005...}

= 1.030199844

Calculates the Liquid Activity Coefficient.

MolFrac Molar Composition Range.

Formula result = 1.0302

[Help on this function](#) OK Cancel

Predicted Pressure can be found from our notes

$$\gamma_i = \left(\frac{Y_i}{X_i} \right) \left(\frac{P_i}{P^{sat}} \right)$$

$$PP = \gamma_1 * X_1 * P_{toluene}^{sat} + \gamma_2 * X_2 * P_{n-Decane}^{sat}$$

[The results are under NRTL]

Experimental			NRTL			
gamma ₁	gamma ₂	ln(gamma ₁) - ln(gamma ₂)	gamma ₁	gamma ₂	Predicted Pressure (kPa)	(Residual) ²
#DIV/0!	1.0000		1.2178	1.0000	9.450000	0.000000
1.0555	0.9676	0.0869	1.2013	1.0116	12.418245	0.621330
1.0303	0.9949	0.0350	1.1858	1.0294	15.303667	1.262628
0.9625	1.0087	-0.0469	1.1639	1.0536	19.468486	4.966151
1.0487	1.0228	0.0250	1.1387	1.0880	24.485831	3.153577
1.0077	1.0133	-0.0056	1.1012	1.1250	32.386668	8.160550
1.0264	0.9860	0.0401	1.0823	1.1445	36.751043	6.205297
1.0222	0.9892	0.0328	1.0445	1.1742	46.949749	2.923240
0.9999	1.0244	-0.0243	1.0297	1.1838	52.003957	3.857126
0.9579	1.2246	-0.2457	1.0099	1.1973	61.664314	8.906129
0.9958	1.2899	-0.2587	1.0038	1.2058	66.776243	0.165034
1.0000	#DIV/0!		1.0000	1.2178	75.380000	0.000000

LEAST-SQUARES SOLVER PANEL	
SSQ	40.22106
A12	0.10000
A21	0.10000
B12	0.00000
B21	0.00000
C	0.30000

Using solver we can get values for A12 and A21 which affect our model’s predicted pressure by setting our objective function as the sum of the Residual column. Set our objective function to a minimum by changing A12, A21, B12, and B21.

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☐ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

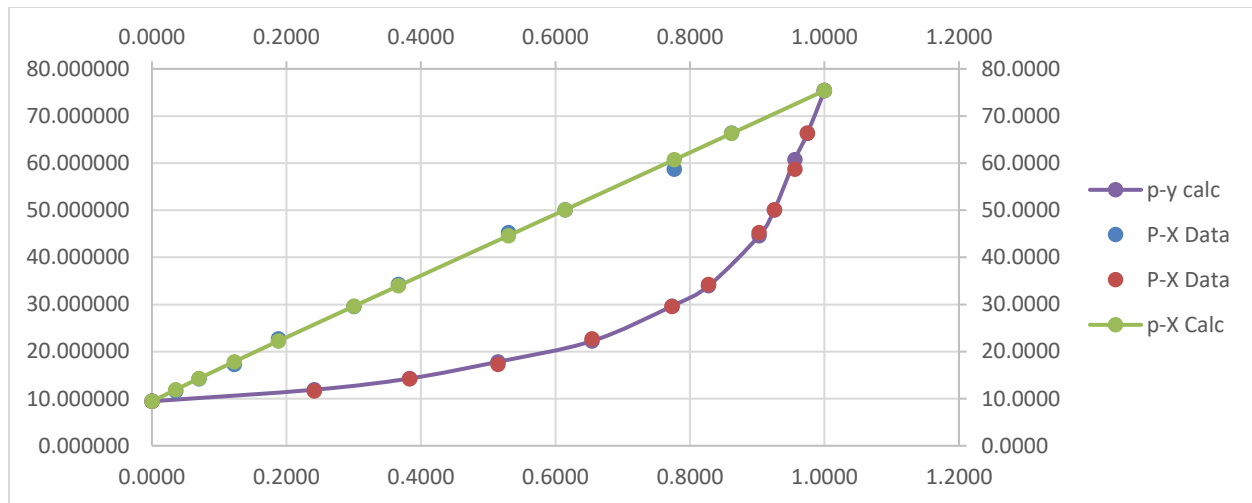
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Help Solve Close

[The final values of my excel sheet]

Experimental			NRTL				LEAST-SQUARES SOLVER PANEL	
γ_1	γ_2	$\ln(\gamma_1) - \ln(\gamma_2)$	γ_1	γ_2	Predicted Pressure (kPa)	(Residual) ²	SSQ	5.38310
#DIV/0!	1.0000		1.0417	1.0000	9.450000	0.000000	A12	-0.48907
1.0555	0.9676	0.0869	1.0366	1.0032	11.903876	0.075008	A21	0.60731
1.0303	0.9949	0.0350	1.0321	1.0067	14.293655	0.012917	B12	-0.09084
0.9625	1.0087	-0.0469	1.0262	1.0102	17.838631	0.358359	B21	0.07872
1.0487	1.0228	0.0250	1.0199	1.0134	22.230072	0.230331	C	0.30000
1.0077	1.0133	-0.0056	1.0119	1.0155	29.641264	0.012380		
1.0264	0.9860	0.0401	1.0085	1.0163	33.946078	0.098547		
1.0222	0.9892	0.0328	1.0032	1.0169	44.600885	0.408468		
0.9999	1.0244	-0.0243	1.0017	1.0170	50.096259	0.003165		
0.9579	1.2246	-0.2457	1.0003	1.0171	60.723833	4.177252		
0.9958	1.2899	-0.2587	1.0001	1.0172	66.288294	0.006676		
1.0000	#DIV/0!		1.0000	1.0172	75.380000	0.000000		

[A graph showing the experimental data versus our calculated]



From earlier we got an equation with $\ln \gamma_2^\infty$ with a relation of only τ_{12} and τ_{21} . Understanding that infinite is when there is a single component, we can get a value for $\ln \gamma_1^\infty$ and $\ln \gamma_2^\infty$ from the experimental data. We have the left side of the equation, now make cells for τ_{12} and τ_{21} . Guess initial values for them. I used .5 and 0 as they were close to the values I wanted.

objective func	1	
Tou 12	0.50000	
Tou 21	0.00000	
	Right Side	Left Side
gam1	0.054014588	0.430354
gam2	0.254564696	0.5

[Here is the solver, I set dummy objective function. With the constraints of gam1 and gam2 having their right and left side equal to each other by changing τ_{12} and τ_{21} .]

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☐ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
 Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

[Final Results]

objective function	1	
Tou 12	1.33776	
Tou 21	-0.84152	
	Right Side	Left Side
gam1	0.054014588	0.054014
gam2	0.254564696	0.254564

Missing minimization of SSQ - 5pts

[Parity plot of predicted pressure values vs. experimental values]

